

Chemistry Chapter 10

Acids, Bases aur Salts

Aasan Notes, tumhari kitab ke saath saath (book pages 140 to 149)

Assalam o alaikum Seemab! Ye chapter class mein ho chuka hai. Ye notes usay **bilkul pakka** karne ke liye hain, bohat aasan alfaaz aur ghar ki misalon ke saath. **Kitab ki har line, har table aur har exercise ka jawab isi mein hai**, kitab kholne ki zaroorat nahin. Headings aur page numbers kitab wale hi hain.

Pichle paper mein jo 3, 4 cheezein thori mushkil lagi thin, un par khaas

 **"Pakki trick"** boxes hain. Unhein zaroor parhna!

Kaise parhna hai (sirf 30 se 35 minute):

1. Ek dafa aaram se parho. Har hisse ke aakhir mein **"Ek line mein"** box hai: bas woh yaad rahe to kaafi hai.
2. Phir neelay  boxes dobara parho: ye kitab ke English alfaaz hain, exam mein yehi likhne hain.
3. Aakhir mein 12 sawal khud karo, jawab agle page par hain.

Is chapter ka naqsha (6 hisse)

 Hissa 1	Acid aur base pehchano (10.1, Table 10.1)
 Hissa 2	Arrhenius concept: paani mein H^+ ya OH^- (10.1.1)
 Hissa 3	Bronsted-Lowry concept: proton dene wala, lene wala (10.1.2)
 Hissa 4	Strong aur weak; alkalis (10.2)
 Hissa 5	Acids aur bases ke kaam (reactions) (10.3, 10.4)
 Hissa 6	Oxides aur hydroxides (10.5)

Student Learning Outcomes (kitab p140): chapter ke baad tumhein ye aana chahiye

- Define Bronsted-Lowry acids as proton donors and Bronsted-Lowry bases as proton acceptors.
- Recognize that aqueous solutions of acids contain H^+ ions and aqueous solutions of alkalis contain OH^- ions.
- Define a strong acid and a base as an acid or base that completely dissociates in aqueous solution, and weak acids and bases that partially dissociate in aqueous solution (for example, writing symbol equations for hydrochloric acid, sulphuric acid, nitric acid and ethanoic acid).
- Formulate dissociation equations for an acid or base in aqueous solution.
- Recognize that bases are oxides or hydroxides of metals and that alkalis are water-soluble bases.
- Describe the characteristic properties of acids in terms of their reactions with metals, bases, and carbonates.
- Identify the characteristic properties of bases in terms of their reactions with acids and ammonium salts.
- Define acid rain.
- Discuss effects of acid rain and relate them with the properties of acids.

Ye sab neeche notes mein aa raha hai.

Hissa 1: Acid aur base pehchano

Introduction (kitab p141)

Acids aur bases hamare ghar mein har jagah hain:

Acids wali cheezein (khatti)	Bases wali cheezein
Vinegar (sirka), lemon juice (nimbu), tomato, apple, cola drinks, aspirin, toilet cleaners	Drain cleaner, antacid tablets, baking powder, soda

- Kitab: "You often use acids and bases in **all areas of life**." Hum kuch acids aur bases **khaate peete** bhi hain, aur **hamara jism** bhi inhein banata hai.
- "In this chapter, you will learn about the chemistry of acids and bases. This will help you better understand these important classes of compounds."
- "Acid indigestion" (tezabiyat) se le kar "acid rain" tak, acid ka lafz **news aur advertisements** mein baar baar aata hai.
- Acids **fertilizers (khaad)** banane mein aur **food industry** mein bohat istemal hote hain.

Kitab ke intro ke sawal (ye chapter inhi ke jawab deta hai): Acid rain kya hai? Kaunsi cheezein acids aur kaunsi bases hain? Un ki classification kaise hoti hai? Acid aur base milein to kya hota hai?

Intro mein do sawal aur bhi hain: "Why do we use lemon juice on fish?" aur "What do we mean by the pH of a solution like acid rain?" Kitab in dono ke jawab is chapter mein nahin deti; pH is chapter mein padhaya nahin gaya, sirf zikr hai.

10.1 Concepts of Acids and Bases (acid aur base ki pehchan, kitab p141)

Kitab: "**Acids and bases are generally recognized by their characteristic properties.**" Yani acid aur base ko un ki khaas properties se pehchante hain.

Table 10.1: Some characteristic properties of acids and bases (poori yaad karo, 5 lines hain bas):

Property	Acid	Base
Taste (zaiqa)	Sour (khatta)	Bitter (kadwa)
Blue litmus par	Turns red	No effect
Red litmus par	No effect	Turns blue
Skin (jild) par	Corrosive	Corrosive
Electrical conductivity	Aqueous solutions conduct electricity	Aqueous solutions conduct electricity

Ghar ki misal

Nimbu aur imli **khatte** hain: acids. Sabun ko zabaan par lagao (mat lagana!) to **kadwa** lagega aur haath mein **phisalta** hai: base.

Litmus ki trick

Base aur **Blue**, dono **B** se! Base red litmus ko **blue** karta hai.
Acid bilkul ulta: blue litmus ko **red** karta hai.

Ek line mein

Acid = khatta, blue ko red karta hai. Base = kadwa, red ko blue karta hai.
Dono corrosive, dono ke solutions bijli guzarte hain.

Hissa 2: Arrhenius concept

10.1.1 Arrhenius Concept of Acids and Bases (kitab p141, 142)

1887 mein Sweden ke chemist **Svante Arrhenius** ne acids aur bases ki pehli kamyab theory di. Ye theory **paani** ke baare mein hai: cheez ko paani mein daalo aur dekho kya nikalta hai.

Kitab ki definition: exam mein yehi likho

"An acid is a substance that ionizes in water to produce H^+ ions and a base is a substance that ionizes in water to produce OH^- ions."

Ionize ka matlab: biscuit todna

Ionize = molecule ka **tootna**, ions mein. Jaise biscuit todo: **poora biscuit pehle** (left side), **tukre baad mein** (right side).

📌 **Pakki trick: equation ULTI mat likhna**

Pichle paper mein equation ulti likh di thi (" $\text{H} + \text{NO}_3 \rightarrow \text{HNO}_3$ "). Hamesha yaad rakho:

Poora molecule LEFT par $\rightarrow$ toote hue ions RIGHT par.

Sahi: $\text{HNO}_3 \rightarrow \text{H}^+ + \text{NO}_3^-$ Ghalat: $\text{H}^+ + \text{NO}_3^- \rightarrow \text{HNO}_3$

Kitab ki equations (arrow ke upar H_2O ka matlab: paani mein):

Cheez	Paani mein tootne ki equation
HCl (acid)	$\text{HCl(g)} \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
HNO_3 (acid)	$\text{HNO}_3(\text{l}) \rightarrow \text{H}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$
H_2SO_4 (acid)	$\text{H}_2\text{SO}_4(\text{l}) \rightarrow 2\text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$
NaOH (base)	$\text{NaOH(s)} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$
KOH (base)	$\text{KOH(s)} \rightarrow \text{K}^+(\text{aq}) + \text{OH}^-(\text{aq})$
NH_4OH (base)	$\text{NH}_4\text{OH(aq)} \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$

H_2SO_4 mein 2 H hain, is liye 2H^+ nikalte hain. Acid se hamesha H^+ , base se hamesha OH^- .

❓ **Kitab ka sawal (p142): "Which substances in the following reactions are acids or bases?"**

HNO_3 aur H_2SO_4 **acids** hain (H^+ dete hain). KOH aur NH_4OH **bases** hain (OH^- dete hain).

Table 10.2: Some Common Acids names, formula and their uses (pehli 4 lines) aur **Table 10.3: Some Common Bases names, formula and their uses** (aakhri 4 lines):

Naam	Formula	Kaam (use)	 Trick
Hydrochloric acid	HCl	Cleaning of metals, bricks and removing scale from boilers	Cl = Clean
Nitric acid	HNO ₃	Fertilizers, explosives	N = Nitrogen khaad
Sulphuric acid	H ₂ SO ₄	Many chemicals, drugs, dyes, paints and explosives	S = Sab kuch
Phosphoric acid	H ₃ PO ₄	Fertilizers, acidulant for food	P = Plants aur Plate (khana)
Sodium hydroxide	NaOH	Soap making, drain cleaners	Na = Nahane ka sabun
Potassium hydroxide	KOH	Liquid soap, shaving cream	K = Kreem
Calcium hydroxide	Ca(OH) ₂	Mortar, plasters, cement	Ca = Construction
Magnesium hydroxide	Mg(OH) ₂	Antacid, laxative	Mg = Medicine

🌟 **Ek line mein**

Arrhenius (1887): paani mein **H⁺ de** = acid, paani mein **OH⁻ de** = base.
Molecule left par, ions right par.



Hissa 3: Bronsted-Lowry concept

10.1.2 The Bronsted-Lowry Concept of Acids and Bases (kitab p142, 143)

Pehle: Arrhenius theory ki kami

Arrhenius theory sirf **paani wale solutions** (aqueous) par chalti hai. Aur ye nahin bata sakti ke:

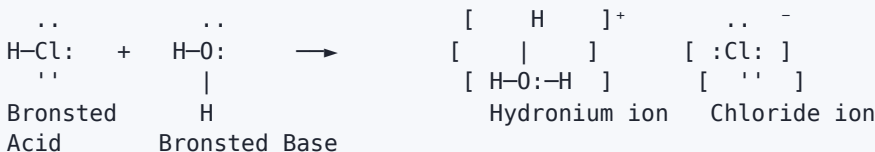
- **CO₂ aur SO₂ acidic kyun hain?** Un mein to koi H hi nahin!
- **NH₃ base kyun hai?** Us mein to koi OH hi nahin!

Is kami ko door karne ke liye **1923** mein **J. N. Bronsted** aur **T. M. Lowry** ne alag alag (independently) nayi theory di.

Kitab ki definition: exam mein yehi likho

“According to this theory, an acid is a proton donor and a base is a proton acceptor.”

Kitab ki pehli misal (p142, dot diagram):



Kitab ki tasveer (dot diagram) mein: HCl ka H gol daire mein hai aur ek mura hua arrow (curved arrow) dikhata hai ke ye H⁺ paani ke O ki taraf jata hai. Paani ka O apne akele electron pair (lone pair) se ye proton pakar leta hai. Is tarah [H₃O]⁺ bracket mein + charge ke saath banta hai (O teen H se juda, ek lone pair baqi), aur Cl⁻ char lone pairs ke saath – charge par reh jata hai.

- HCl apna H⁺ (proton) paani ko de deta hai: **HCl = Bronsted acid.**
- Paani proton le leta hai: **H₂O = Bronsted base.**
- Banne wali cheezein: **H₃O⁺ = hydronium ion, Cl⁻ = chloride ion.**
(Hydronium ka naam yaad rakho!)

? Kitab ke 4 sawal is misal par (p143)

- Q1. Which substance is donating proton? **HCl**
Q2. Which substance is accepting proton? **H₂O**
Q3. Which substance is an acid? **HCl**
Q4. Which substance is a base? **H₂O**

🍫 Chocolate wali misal (sab se aasan)

Proton = H⁺ = ek chocolate.

Jo chocolate **DETA** hai (donor) woh **ACID**.

Jo chocolate **LETA** hai (acceptor) woh **BASE**.

Bas itni si baat hai!

🎯 Pakki trick: H gino!

Pichle paper mein kabhi paani ka role ulta ho gaya, kabhi H₃O⁺ ko acid likh diya. Ab ye 3 steps karo:

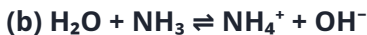
1. Reaction ka acid aur base hamesha **LEFT side** (reactants) mein se chuno. Right side wali cheezein (H₃O⁺, Cl⁻, NH₄⁺, OH⁻) banne wali cheezein (products) hain, woh **is reaction** ka acid ya base nahin. (Khud woh bhi proton de ya le sakti hain: H₃O⁺ aur NH₄⁺ proton de sakte hain, OH⁻ aur Cl⁻ le sakte hain. Lekin sawal hamesha reaction ke acid/base ka hota hai.)
2. Left wali cheez ka right side par kya bana? **H kam hua** (chocolate de di) → woh **ACID**.
3. **H barh gaya** (chocolate mil gayi) → woh **BASE**.

 Example 10.1 (kitab p143): "Classify substances as acids or bases or as proton donor or proton acceptor"



$\text{HCl} \rightarrow \text{Cl}^-$ H kam hua, chocolate di **$\text{HCl} = \text{acid}$**

$\text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+$ H barha, chocolate li **$\text{H}_2\text{O} = \text{base}$**

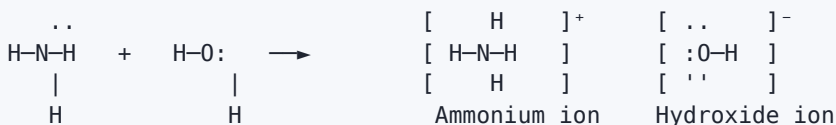


$\text{H}_2\text{O} \rightarrow \text{OH}^-$ H kam hua, chocolate di **$\text{H}_2\text{O} = \text{acid}$**

$\text{NH}_3 \rightarrow \text{NH}_4^+$ H barha, chocolate li **$\text{NH}_3 = \text{base}$**

Kitab ka solution: 1. Because HCl is converted to Cl^- by donating proton, HCl is an acid. 2. Because H_2O accepts the proton that HCl donates and forms H_3O^+ , water is a base. 3. H_2O is converted to OH^- by donating a proton, so H_2O is an acid. Because NH_3 accepts the proton and forms NH_4^+ , so it is a base.

Ammonia wali misal (kitab p143): Arrhenius theory nahin bata sakti ke ammonia (NH_3) base kaise hai, kyunke us mein OH hai hi nahin. Bronsted-Lowry theory batati hai. **Ammonia room temperature par gas hai.** Paani mein ghulne par:



Kitab ki tasveer mein: NH_3 ke N par ek lone pair hai. Mura hua arrow N ke lone pair se paani ke gol daire wale H ki taraf jata hai: NH_3 paani ka H^+ le leta hai. Is se $[\text{NH}_4]^+$ (bracket, + charge) aur $[\text{:O}-\text{H}]^-$ (bracket, - charge) bante hain.

Paani apna ek H^+ NH_3 ko deta hai. NH_3 ban jata hai **NH_4^+ (ammonium ion)** aur paani ban jata hai **OH^- (hydroxide ion)**. Is liye yahan **$\text{NH}_3 = \text{base}$** (proton acceptor) aur **$\text{H}_2\text{O} = \text{acid}$** (proton donor).

★ Kitab ki do lines (p143)

"All the acids included in the Arrhenius theory are also acids in the Bronsted-Lowry theory."

"However, all the bases included in Bronsted-Lowry theory except OH^- are not Arrhenius bases."

Matlab: har Arrhenius acid Bronsted acid bhi hai. Lekin Bronsted bases (jaise NH_3) Arrhenius bases nahin, kyunke un mein OH nahin hota. **Sirf OH^- dono theories mein base hai:** Arrhenius base OH^- deta hai, aur OH^- khud proton le leta hai (Bronsted base).

Dekha? **Paani (a) mein base tha aur (b) mein acid!** Isi liye kitab kehti hai: water behaves like an acid as well as a base. It is **amphoteric** in nature.

📖 Kitab ki definition

"Substances that react with both acids and bases are called **amphoteric substances**."

👥 Paani ki misal

Paani us saheli jaisa hai jo HCl ke saath ho to chocolate **le leti** hai (base), aur NH_3 ke saath ho to chocolate **de deti** hai (acid). Saathi dekh kar role badalta hai!

★ Kitab ki problem solving strategy (p143)

1. Acid proton de kar **negative ion** ban jata hai ($\text{HCl} \rightarrow \text{Cl}^-$).
2. Base proton le kar **positive ion** ban jata hai ($\text{NH}_3 \rightarrow \text{NH}_4^+$, $\text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+$).
Aur yaad rakho: Arrhenius wale **sab acids** Bronsted-Lowry mein bhi acids hain.

🔥 Concept Assessment Exercise 10.1 (kitab p144): jawab ke saath

1. $\text{H}_2\text{SO}_4 + \text{H}_2\text{O} \rightarrow \text{HSO}_4^- + \text{H}_3\text{O}^+$ **H_2SO_4 acid, H_2O base**
2. $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$ **CH_3COOH acid, H_2O base**
3. $\text{H}_2\text{S} + \text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{HS}^-$ **H_2S acid, NH_3 base**

Har ek mein trick lagao: kis ka H kam hua, kis ka barha?

🌟 Ek line mein

Bronsted-Lowry (1923): acid = **proton donor** (H deta hai), base = **proton acceptor** (H leta hai). Paani dono ban sakta hai: **amphoteric**.

💪 Hissa 4: Strong, weak aur alkalis

10.2 Strength of acids and bases (kitab p144)

Kitab: "**All the acids and bases do not ionize in aqueous solutions to the same degree. Therefore, they have different strengths.**" Yani sab paani mein ek jitna nahin toot-te, is liye kuch strong hain, kuch weak. Bases bhi acids ki tarah alag alag had tak ionize hote hain.

📖 Kitab ki definitions

"An acid that ionizes completely in aqueous solution is called a strong acid."

"Acids that do not ionize completely in aqueous solutions are called weak acids."

"A base that ionizes completely or 100% in aqueous solution is termed as a strong base."

"A base that ionizes to a little extent is called a weak base."

	Strong (100% toot-ta)	Weak (thora sa toot-ta)
Acids	HCl, HNO ₃ , H ₂ SO ₄	CH ₃ COOH (ethanoic / acetic acid, sirke mein): sirf 5% tak
Bases	NaOH, KOH, Ca(OH) ₂	Al(OH) ₃ , NH ₃ (kam OH ⁻ dete hain)
Arrow	Ek arrow →	Do arrow ⇌

10.2.1 aur 10.2.2 ki equations (kitab p144):

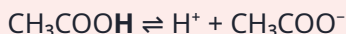
10.2.1 Strong and weak acids: strong (sab molecules ionize, 100%)	10.2.2 Strong and weak bases: strong (100%)
$\text{HCl(g)} \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$ $\text{HNO}_3(\text{l}) \rightarrow \text{H}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$ $\text{H}_2\text{SO}_4(\text{l}) \rightarrow 2\text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$	$\text{NaOH(s)} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$ $\text{KOH(s)} \rightarrow \text{K}^+(\text{aq}) + \text{OH}^-(\text{aq})$
Weak acid (kam molecules ionize)	Weak base (kam OH^- deta hai)
$\text{CH}_3\text{COOH} \rightleftharpoons \text{H}^+ + \text{CH}_3\text{COO}^-$	$\text{H}_2\text{O} + \text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$

Biscuit packet ki misal

Strong acid = poore packet ke **sab** biscuit toot gaye (100%).

Weak acid = 100 biscuits mein se sirf **5** toote (sirke wala acetic acid).

Pakki trick: CH_3COOH ka sirf AAKHRI H alag hota hai



CH_3 wale H kabhi alag nahin hote. Toota hua hissa poora likho: CH_3COO^- (CH_3 mat bhoolna). Aur weak hai, is liye $\rightleftharpoons$ lagao.

Alkalis (kitab p145)

Kitab ki definition

"A base which is soluble in water is called an alkali."

- **Sab alkalis bases hain, lekin sab bases alkalis nahin.**
- Paani mein na gholne wale bases: $\text{Cu}(\text{OH})_2$ (copper hydroxide), $\text{Al}(\text{OH})_3$ (aluminium hydroxide), $\text{Fe}(\text{OH})_3$ (ferric hydroxide).
- **"All hydroxides are bases and only water soluble bases are alkalis."**
Bohat se bases paani mein nahin ghulte.
- Ghar ke alkalis: **soaps, detergents, shampoos, toothpaste.**
- KOH ya NaOH, kaun alkali? **Dono**, kyunke dono paani mein ghul jate hain.

Aasan misal

Har Lahori Pakistani hai, lekin har Pakistani Lahori nahin. Isi tarah har **alkali** base hai, lekin har base alkali nahin!

Ek line mein

Strong = poora toot-ta ($\rightarrow$), weak = thora sa ($\rightleftharpoons$). Paani mein ghulne wala base = **alkali**.



Hissa 5: Acids aur bases ke kaam

10.3 Characteristic properties of acids (kitab p145, 146)

Acids ki 5 pehchan:

1. Acids have **sour taste**.
2. Acids change the colour of **blue litmus paper to red**.
3. **Metals ke saath:** "Acids react with most metals and corrode them." Zinc, magnesium, aluminium aur calcium jaise metals ke saath un ka salt aur **hydrogen gas** banti hai, jo **bulbulon** ki shakal mein nikalti hai. Zinc + hydrochloric acid $\rightarrow$ **zinc chloride** + hydrogen. Magnesium + sulphuric acid $\rightarrow$ **magnesium sulphate** + hydrogen.
4. **Metal carbonates ke saath:** salt + water + **carbon dioxide**, jo paani mein bulbule banati hai. Hydrochloric acid + sodium carbonate $\rightarrow$ **sodium chloride** + carbon dioxide + water. Sulphuric acid + calcium carbonate (limestone ya marble) $\rightarrow$ **calcium sulphate** + carbon dioxide + water.
5. **Bases ke saath:** salt + water. Isay **neutralization reaction** kehte hain. Hydrochloric acid + sodium hydroxide $\rightarrow$ **sodium chloride** + water. Sulphuric acid + potassium hydroxide $\rightarrow$ **potassium sulphate** + water.

 Kitab ke alfaaz (p145): acids ki properties 3, 4, 5

3. "Acids react with most metals and corrode them. Acids combine with metals like zinc, magnesium, aluminium and calcium to form their salts and hydrogen gas. The hydrogen gas is liberated in the form of bubbles." Metal + Acid → Salt + Hydrogen gas

4. "Acids react with metal carbonates to form salts, water with the liberation of carbon dioxide. The liberated carbon dioxide forms bubbles in water." Metal carbonate + Acid → Salt + Water + Carbon dioxide. **"This reaction is used in the industrial preparation of glass, paper and soap."**

5. "Acids combine with bases to produce salt and water. This reaction is called as neutralization reaction." Acid + Base → Salt + Water

 Teen reactions, ek trick: "Salt har jagah!"

Acid + **Metal** → Salt + **Hydrogen**

Acid + **Carbonate** → Salt + Water + **CO₂** (Carbonate se CO₂)

Acid + **Base** → Salt + **Water** (neutralization)

 Pakki trick: "2" lagana mat bhoolna (kitab bhi bhool gayi!)

Kitab ne 3 misalon mein balance karna chhor diya hai. **Tum sahi wali likhna:**

Kitab mein chhapa (ghalat) ✗	Sahi, balanced ✓
$\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$
$\text{H}_2\text{SO}_4 + \text{KOH} \rightarrow \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$	$\text{H}_2\text{SO}_4 + 2\text{KOH} \rightarrow \text{K}_2\text{SO}_4 + 2\text{H}_2\text{O}$
$\text{H}_2\text{SO}_4 + \text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$	$\text{H}_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$



Balance kaise karein: atoms gino

$\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ (abhi balanced nahin)

Na: left 2, right 1 → right par **2NaCl**

Cl: ab right 2, left 1 → left par **2HCl**

H: left 2, right 2 **barabar**

C, O: C 1 = 1, O 3 = 3 **barabar**

$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ Done!

Aakhir mein har atom ek ek kar ke gin lo. 30 second lagte hain aur poore marks milte hain.

Kitab ki baqi misalein (ye pehle se balanced hain):

Reaction	Equation
Acid + metal	$\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
Acid + metal	$\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$
Acid + carbonate	$\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$ CaCO_3 = limestone ya marble. Acid + carbonate wali reaction glass, paper aur soap banane mein kaam aati hai.
Neutralization	$\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$



Neutralization ki ghar wali misal

Pet mein tezabiyat (acidity)? Doctor **antacid** deta hai. Antacid mein base hota hai, jaise **$\text{Mg}(\text{OH})_2$** (milk of magnesia). Ye pet ke acid (HCl) ko **neutralize** kar deta hai: acid + base → salt + water. Aag par paani jaisa!

Acid rain (kitab p145, 146):

 **Kitab ke alfaaz: exam mein yehi likho (SLO: Define acid rain, effects of acid rain)**

“Normal rain is slightly acidic due to dissolved carbon dioxide. As a result of human activity, many oxides enter the atmosphere, which makes rainwater more acidic, which falls as acid rain. Acid rain contains dissolved nitric and sulphuric acid. Due to the corrosive nature of acids, acid rain can damage structures, buildings, and statues containing metals and metal carbonates.”

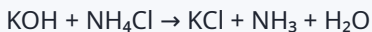
- **Normal baarish** bhi thori si acidic hoti hai, kyunke us mein **carbon dioxide** ghuli hoti hai.
- Insaan ki activities se **bohat se oxides** hawa mein jaate hain aur baarish ka paani **zyada acidic** ho jata hai: ye **acid rain** hai.
- Acid rain mein **nitric acid** aur **sulphuric acid** ghule hote hain.
- Acids **corrosive** hain, is liye acid rain **metals aur metal carbonates** wali buildings, structures aur statues ko nuqsan deti hai. (Is ki poori kahani Chapter 11 mein hai.)

10.4 Characteristic properties of bases (kitab p146)

1. Bases have **bitter taste**.
2. Bases change the colour of **red litmus paper to blue**.
3. Aqueous solution of bases have **slippery touch**. (Sabun wala paani haath par phisalta hai!)
4. Bases **neutralize acids** to form salt and water: $\text{H}_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
5. **"Bases decompose ammonium salts on heating and liberate ammonia gas."** (Garam karne par base ammonium salt ko tod deta hai aur ammonia gas nikalti hai.)

Base + Ammonium salt → Salt + Ammonia + Water

$\text{NaOH} + \text{NH}_4\text{Cl}$ deta hai **sodium chloride**, $\text{KOH} + \text{NH}_4\text{Cl}$ deta hai **potassium chloride**, dono mein ammonia gas aur paani.



🔗 Pakki trick: gas ka NAAM bhi likho

Sirf "NH₃" likhna kaafi nahin. Jab sawal gas poochhe to alag se likho:

"The gas released is ammonia (NH₃)."

Yaad kaise rakhein? **Ammonium** salt se **Ammonia** gas. Naam mein hi likha hai!

✨ Ek line mein

Acid + metal = H₂; acid + carbonate = CO₂; acid + base = salt + water (neutralization). Base + ammonium salt = ammonia gas. Balance karna aur gas ka naam, dono mat bhoolo.



Hissa 6: Oxides aur hydroxides

10.5 Oxides and hydroxides (kitab p146)

Bases = metals ke oxides aur hydroxides.

📖 Kitab ki definition

"Metallic oxides are the compounds formed by the reaction of metals with oxygen."

Misal: sodium oxygen se react kar ke **sodium oxide (Na₂O)** banata hai.

Magnesium hawa mein aag lagne par (on ignition) jal kar **magnesium oxide (MgO)** banata hai.

Metal + Oxygen → Metal oxide

Equation	Naam (English mein bhi likho!)
$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$	Na ₂ O = sodium oxide
$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$	MgO = magnesium oxide

🎯 **Pakki trick: oxygen hamesha JORI mein**

Oxygen gas hamesha **O₂** likhi jati hai, jaise jutay hamesha jori mein. Akeli "O" kabhi nahin.

Naam ka formula: **metal ka naam + "oxide"**. Na₂O = sodium oxide, MgO = magnesium oxide, CaO = calcium oxide.

Metal oxide + paani → metal hydroxide. Kitab ke alfaaz: metal oxide paani mein ghul kar **metal cations** aur **oxide ions** deta hai. **Oxide ions paani mein unstable hote hain**, is liye foran paani ke molecules se proton (H⁺) le kar **hydroxide ions** ban jate hain, aur paani ke molecules bhi hydroxide ions ban jate hain. Yani oxide ion **Bronsted base** hai.

❓ **Kitab ka sawal (p146): "Which species is the proton donor in the following reaction?"**

Water (H-OH) proton deta hai, is liye paani proton donor hai.



Equation	Naam
$\text{MgO} + \text{H}_2\text{O} \rightarrow \text{Mg(OH)}_2$	magnesium hydroxide
$\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$	sodium hydroxide
$\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$	calcium hydroxide

- **"Most metal oxides and hydroxides are very basic in nature. They show the characteristic properties of bases."**
- Lekin kuch metal oxides aur hydroxides paani mein nahin ghulte aur chemical reactions mein **acid aur base dono** jaisa kaam karte hain. Inhein **amphoteric oxides** aur **amphoteric hydroxides** kehte hain.
- Misalein: **aluminium oxide (Al₂O₃)**, **aluminium hydroxide (Al(OH)₃)**, **zinc oxide (ZnO)**, **zinc hydroxide (Zn(OH)₂)**.

 **Kitab ke alfaaz (p146): exam mein yehi likho**

"Some metal oxides and hydroxides do not dissolve in water but behave in chemical reactions like both acids and bases. Such metal oxides and hydroxides are called amphoteric oxides and hydroxides, respectively."

 **Amphoteric ki trick: "A se Z tak"**

Aluminium aur Zinc: A se Z, yani **dono taraf!** In ke oxides aur hydroxides acid aur base dono ke saath react karte hain.

 **Ek line mein**

Metal + $O_2 \rightarrow$ metal oxide; metal oxide + paani $\rightarrow$ metal hydroxide (base).
Aluminium aur zinc wale = amphoteric.

Key Points: kitab ke alfaaz mein

- According to Arrhenius theory, an acid is a substance that ionizes in water to produce H^+ ions and a base is a substance that ionizes in water to produce OH^- ions.
- A Bronsted-Lowry acid is a proton donor and a base is a proton acceptor.
- A strong acid completely dissociates in aqueous solution and a weak acid dissociates partially in aqueous solution.
- A strong base completely dissociates in aqueous solution and a weak base dissociates partially in aqueous solution.
- Bases are oxides and hydroxides of metals.
- Alkalis are water soluble bases.
- Acids neutralize bases to form salt and water.
- Acids decompose carbonates to form salt, water, and carbon dioxide.
- Acids corrode metals and form salt and hydrogen.

References for additional information (kitab p147): Longman Chemistry for IGCSE; IGCSE Chemistry; Cambridge IGCSE Chemistry; Theories of Acids and Base, Chemguide. (Ye sirf mazeed parhne ke liye kitabein hain, exam mein nahin aatein.)

 Kitab ke Review Questions mein 2 ghaltiyan (p148, Q4)

(ii) "Sulphuric acid + Sodium carbonate → Sodium sulphate + water":
carbon dioxide bhi banti hai. Sahi: $\text{H}_2\text{SO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O} + \text{CO}_2$

(iii) "Hydrochloric acid + Magnesium → Magnesium chloride + water":
water nahin, **hydrogen** banti hai. Sahi: $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$



Kitab ke Review Questions: sab ke jawab (p147 to 149)

Q1. Encircle the correct answer (MCQs)

Sawal	Options	Jawab
(i) Which of the following cannot be classified as Arrhenius acid?	(a) HNO_3 (b) H_2CO_3 (c) CO_2 (d) H_2SO_4	(c) CO_2: is mein H hai hi nahin
(ii) Which of the following is a Bronsted base?	(a) NH_3 (b) HCl (c) CH_3COOH (d) H_3O^+	(a) NH_3: proton leta hai
(iii) Milk of magnesia contains $\text{Mg}(\text{OH})_2$. It is used as antacid. It neutralizes excess stomach acid. Which salt is formed in this reaction?	(a) MgSO_4 (b) MgCO_3 (c) MgCl_2 (d) MgO	(c) MgCl_2: stomach acid HCl hai, $\text{Mg}(\text{OH})_2 + 2\text{HCl} \rightarrow \text{MgCl}_2 + 2\text{H}_2\text{O}$
(iv) Ammonia is a base, because it	(a) ionizes in water to give OH^- ions (b) contains OH group (c) can accept an electron pair (d) can accept proton	(d) can accept proton (Bronsted-Lowry)
(v) $\text{H}_2\text{O} + \text{HCl} \rightleftharpoons \text{H}_3\text{O}^+ + \text{Cl}^-$. Which species is a proton acceptor in this reaction? (kitab mein "electron proton acceptor" chhapa hai; "electron" ghalti se hai)	(a) H_2O (b) HCl (c) H_3O^+ (d) none	(a) H_2O: H barha, H_3O^+ bana

Q2. Give short answer

i. Write the equation for the self-ionization of water.

$\text{H}_2\text{O} + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{OH}^-$ (ek paani molecule proton deta hai, doosra leta hai; paani amphoteric hai)

ii. Define and give examples of Arrhenius acids.

An Arrhenius acid is a substance that ionizes in water to produce H^+ ions.

Examples: HCl, HNO₃, H₂SO₄, H₃PO₄.

iii. **Why HCl acts as a strong acid?**

Because it ionizes completely (100%) in aqueous solution: $\text{HCl(g)} \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$.

iv. **Why NH₃ acts as Bronsted-Lowry base?**

Because it accepts a proton from water and forms NH₄⁺: $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$.

v. **Why ammonia acts as a weak base?**

Because it ionizes only to a little extent in water and produces fewer OH⁻ ions: $\text{H}_2\text{O} + \text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$.

Q3. Ammonium hydroxide and nitric acid react and produce ammonium nitrate and water. Write the balanced equation.



Q4. Write balanced chemical equations (kitab ki do ghaltiyan theek kar ke):

(i) Sulphuric acid + magnesium hydroxide → magnesium sulphate + water	$\text{H}_2\text{SO}_4 + \text{Mg(OH)}_2 \rightarrow \text{MgSO}_4 + 2\text{H}_2\text{O}$
(ii) Sulphuric acid + sodium carbonate → sodium sulphate + water + carbon dioxide	$\text{H}_2\text{SO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O} + \text{CO}_2$
(iii) Hydrochloric acid + magnesium → magnesium chloride + hydrogen	$2\text{HCl} + \text{Mg} \rightarrow \text{MgCl}_2 + \text{H}_2$

Q5. Identify Bronsted-Lowry acids or bases (H gino trick):

(i) $\text{HNO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{NO}_3^-$	HNO₃ acid , H ₂ O base
(ii) $\text{NH}_3 + \text{HNO}_3 \rightarrow \text{NH}_4\text{NO}_3$	HNO₃ acid (H deta hai), NH₃ base (H leta hai, NH ₄ ⁺ banta hai)

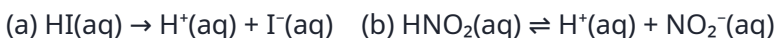
Q6. Give the Bronsted-Lowry definition of an acid. Write an equation that illustrates the definition.

An acid is a proton donor. Example: $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$ (HCl donates a proton to water).

Q7. Identify Bronsted acids and Bronsted bases. Classify water as proton donor or proton acceptor.

Reaction	Acid	Base	Paani
(i) $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$	CH_3COOH	H_2O	acceptor
(ii) $\text{HCO}_3^- + \text{H}_2\text{O} \rightleftharpoons \text{CO}_3^{2-} + \text{H}_3\text{O}^+$	HCO_3^-	H_2O	acceptor
(iii) $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$	H_2O	NH_3	donor
(iv) $\text{HCl} + \text{HCO}_3^- \rightleftharpoons \text{H}_2\text{CO}_3 + \text{Cl}^-$	HCl	HCO_3^-	(paani nahin hai)
(v) $\text{HS}^- + \text{H}_2\text{O} \rightleftharpoons \text{S}^{2-} + \text{H}_3\text{O}^+$	HS^-	H_2O	acceptor

Q8. Write equations showing the ionization of the following as Arrhenius acids.



HI strong acid hai ($\rightarrow$). HNO_2 (nitrous acid) weak acid hai, is liye $\rightleftharpoons$.

Q9 mein kitab ne yehi heading dobara chhaap di hai aur neeche koi sawal nahin; ye printing galti hai, us mein karne ko kuch nahin.

Think Tank

10. Compare the relative concentrations of hydrogen ions and hydroxide ions in each kind of solution.

(a) Acidic: H^+ zyada, OH^- kam. (b) Basic: OH^- zyada, H^+ kam. (c) Neutral: H^+ aur OH^- barabar.

11. Codeine, $\text{C}_{18}\text{H}_{21}\text{NO}_3$, is a commonly prescribed pain killer. It dissolves in water: $\text{C}_{18}\text{H}_{21}\text{NO}_3 + \text{H}_2\text{O} \rightleftharpoons [\text{C}_{18}\text{H}_{21}\text{HNO}_3]^+ + \text{OH}^-$. Differentiate codeine and water as Bronsted-Lowry acid or base.

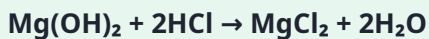
Codeine proton leta hai (positive ion banta hai): **codeine = base**. Paani proton deta hai (OH^- banta hai): **water = acid**.

12. Examine some ways in which you might determine whether a particular water solution contains an acid or a base.

Litmus se: blue litmus red ho jaye to **acid**; red litmus blue ho jaye to **base**. Acid mein zinc ya magnesium daalo to hydrogen ke bulbule; carbonate daalo to CO_2 ke bulbule. Base ka solution haath par phisalta hai (lekin chhuna nahin, corrosive hai).

Project (kitab p149)

Kam az kam teen antacids ke labels dekho, har ek ke ingredients ki list banao, aur pet ke HCl ke saath neutralization ki balanced equation likho. Misal, agar antacid mein magnesium hydroxide ho:





Poora chapter, ek page mein

Acid	Khatta, blue litmus → red, H^+ deta hai (Arrhenius), proton deta hai (Bronsted)
Base	Kadwa, phisalne wala, red litmus → blue, OH^- deta hai (Arrhenius), proton leta hai (Bronsted)
Amphoteric	Acid aur base dono: paani; Al aur Zn ke oxides/hydroxides
Strong	100% ionize (→): HCl, HNO_3 , H_2SO_4 ; NaOH, KOH, $Ca(OH)_2$
Weak	Thora sa ionize ($\rightleftharpoons$): CH_3COOH (5%); $Al(OH)_3$, NH_3
Alkali	Paani mein ghulne wala base (soap, shampoo, toothpaste)
Acid + metal	Salt + H_2
Acid + carbonate	Salt + H_2O + CO_2
Acid + base	Salt + H_2O (neutralization)
Base + NH_4Cl	Salt + NH_3 (ammonia) + H_2O
Metal + O_2	Metal oxide; + H_2O → metal hydroxide
Acid rain	Nitric + sulphuric acid wali baarish; metals aur carbonates ko khaati hai

Paper se pehle 4 pakki baatein

1. Ionization: molecule **left**, ions **right**.
2. Bronsted: **H gino**; sirf left wali cheezon ko acid/base kaho.
3. Equation ke baad **atoms gino**, "2" lagao; oxygen hamesha **O_2** .
4. Gas ka aur banne wali cheez ka **naam English mein** bhi likho.

✓ Khud ko check karo (12 sawal)

Pehle notebook mein khud jawab likho, phir agle page par check karo.

1. Base red litmus ko kis rang ka karta hai?
2. Arrhenius ke mutabiq acid paani mein kaunsa ion deta hai?
3. HCl ke paani mein ionize hone ki equation likho (state symbols ke saath).
4. Bronsted-Lowry acid ki definition likho.
5. $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$ mein Bronsted acid aur base batao.
6. $\text{HNO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{NO}_3^-$: is reaction ka acid kaun hai, HNO_3 ya H_3O^+ ? (Soch kar!)
7. Sirke mein kaunsa acid hai, aur woh strong hai ya weak?
8. Alkali kya hota hai? Ek aisa base batao jo alkali nahin.
9. Balance karo: $\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$
10. $\text{NaOH} + \text{NH}_4\text{Cl}$ ko garam karo to kaunsi gas nikalti hai?
11. $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$: MgO ka English naam kya hai?
12. Do amphoteric oxides ke naam batao.

🧑 Agar kuch samajh na aaye (optional)

NBF Chemistry 9 Chapter 10 exercise ki video: youtube.com/watch?v=zELXCuDYFWI

Notes kaafi hain; video sirf tab dekho jab koi hissa mushkil lage.

Jawab

1. **Blue** (B se Base, B se Blue)
2. **H⁺ ion**
3. **HCl(g) → H⁺(aq) + Cl⁻(aq)** (molecule left, ions right)
4. **An acid is a proton donor.**
5. $\text{NH}_3 \rightarrow \text{NH}_4^+$ (H barha): **NH₃ = base**. $\text{H}_2\text{O} \rightarrow \text{OH}^-$ (H kam hua): **H₂O = acid**.
6. **HNO₃** (left side par, H kam hua). Base **H₂O** hai (H barha). **H₃O⁺** yahan product hai, is reaction ka acid nahin.
7. **Ethanoic acid (acetic acid), CH₃COOH**. Ye **weak** hai, sirf 5% tak ionize hota hai.
8. Paani mein ghulne wala base. Jo alkali nahin: **Cu(OH)₂, Al(OH)₃ ya Fe(OH)₃** (koi ek).
9. **Na₂CO₃ + 2HCl → 2NaCl + H₂O + CO₂**
10. **Ammonia gas (NH₃)**
11. **Magnesium oxide**
12. **Aluminium oxide (Al₂O₃) aur zinc oxide (ZnO)**

Score dekho:

- **10 se 12 sahi:** zabardast! 🌟 Chapter 10 tumhara pakka ho gaya.
- **7 se 9 sahi:** bohat achha! Jo ghalat hue, sirf un ka "Pakki trick" box dobara dekh lo.
- **7 se kam:** koi baat nahin. Kal ek dafa aur parho, doosri dafa hamesha aasan lagta hai. 💪